

FLUIDS PRACTICE FRQ SOLUTIONS

2009B5.

- a) There are three forces acting on the masses in each case. Tension up, buoyant force up, weight down. Since they are at rest we have. $F_{\text{net}} = 0$ $F_t + F_b = mg$ $F_t = mg - F_b$ so the largest F_b makes the largest F_t

We are to assume the diagram is to scale and that clearly the volumes of the three containers are different. The one with the largest volume displaces the largest amount and weight of water and will have the largest buoyant force acting on it. So since they all displace different volumes (and weights) of water they all have different buoyant forces, and based on the equation shown above will have different tensions.

- b) The mass of the object is given by $m = \rho_{\text{obj}} V_{\text{obj}}$.

Using the equation from part a,

$$F_t + F_b = mg, \quad F_t + F_b = (\rho_{\text{obj}} V_{\text{obj}}) g \quad (0.0098) + F_b = (1300)(1 \times 10^{-5})(9.8) \quad F_b = 0.1176 \text{ N}$$

- c) The buoyant force is by definition equal to the weight of the displaced fluid.

$$F_b = (\rho_{\text{fluid}} V_{\text{disp}}) g \quad 0.1176 = \rho_{\text{fluid}} (1 \times 10^{-5})(9.8) \quad \rho_{\text{fluid}} = 1200 \text{ kg/m}^3$$

- d) With only half of the volume submerged, $\frac{1}{2}$ as much water will be displaced and the buoyant force will be half the size. Based on the formula from part A, less buoyant from will make a larger tension. This also makes sense conceptually. Objects have large apparent weights in air than water so having some of it in the air will increase its apparent weight.

1-

2007B4.

- a) Volume flow rate = $Q = V/t = 7.2 \times 10^{-4} / (2 \text{ min} * 60 \text{ sec/min}) = 6 \times 10^{-6} \text{ m}^3/\text{s}$

- b) Your first thought is probably Bernoulli, but there are too many unknowns so this does not work. We can use the volume flow rate above to find the velocity.

$$Q = Av \quad 6 \times 10^{-6} = (2.5 \times 10^{-6}) v \quad v = 2.4 \text{ m/s}$$

- c) Use Bernoulli, same derivation as in the problem above (B2005B5) ... $v_2 = \sqrt{2gh}$ $(2.4) = \sqrt{2(9.8)h}$ $h = 0.29 \text{ m}$

- d) Left of beaker. Based on the formula derived above, the exit velocity is dependent on the height and with less horizontal exit velocity the range will be less ($d_x = v_x t$). This makes sense because less height would result in less pressure and decrease the speed the fluid is ejected at, thus lessening the range.

2-

B2009B3.

a) Using fluid continuity. $A_1 v_1 = A_2 v_2$ $(1 \times 10^{-4})(v_1) = (0.5 \times 10^{-4})(8.2)$ $v_a = 4.1 \text{ m/s}$

b) Bernoulli. Position B is the fountain spigot which is open so at atmospheric pressure. $y_1 = 0$ no height.

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 + 0 + \frac{1}{2} (1000)(4.1)^2 = (1.01 \times 10^5) + (1000)(9.8)(0.5 \text{ m}) + \frac{1}{2} (1000)(8.2)^2$$

$$P_1 = 1.3 \times 10^5 \text{ Pa}$$

c) Free fall of a water droplet. $v_f^2 = v_i^2 + 2gd$ $(0)^2 = (8.2)^2 + 2(-9.8)(d)$ $d = 3.43 \text{ m}$

d) Projectile method, in y direction.

$$d_y = v_{iy}t + \frac{1}{2}gt^2$$

$$d_y = (v_i \sin \theta)t + \frac{1}{2}gt^2$$

$$0 = (8.2 \sin 60)t + \frac{1}{2}(-9.8)t^2$$

$$t = 1.45 \text{ sec}$$

X direction. $d_x = v_x t$ $d_x = (v_i \cos \theta)t$ $d_x = (8.2 \cos 60)(1.45)$ $d_x = 5.95 \text{ m}$